

obvious that we have tested and made it as fail-safe as possible."

Netra can be operated during the day and at night. This is made possible with a 10x zoom payload for daylight and a thermal imaging camera for night operations. The payload itself can tilt while the aircraft can pan the camera to get a full 360-degree view.

So how did the team solve the reliability issues most often encountered with systems of this nature? Rahul Singh, co-founder and CTO, responds, "We attribute the reliability of the system to the use of highly advanced, innovative and integrated inertial sensors from Analog Devices that offer 10 degrees of freedom. The reliability worries were also addressed with an all-in-one device which included a three-axis gyroscope, an accelerometer, a magnetometer and a pressure sensor. As we know, gyros, accelerometers and magnetometers are essential technologies to maintain orientation and track position, tilt, direction and rotation along all the three axes with

dynamic feedback to provide accurate motion measurements—a must for a superior and reliable autopilot. These sensors were invaluable when furnishing the final working prototype."

The challenges faced

Developing an indigenous solution usually involves facing uniquely Indian challenges that, in turn, require indigenous solutions. Speaking about the challenges, Amardeep Singh, chief marketing officer, shares, "While we started work on the quad-rotor when we were in college, we took it up seriously only in 2008. After that, it took us about a year-and-a-half to deliver our first commercial system. Developing an aerial platform requires multidisciplinary skills that are difficult to master. It has been through a lot of experimentation and analysis that we have been able to ensure that we offer a reliable and competitive product. The communication range, endurance and payload weight are typically the most common yet also the most dif-

icult problems to crack in a UAV. But thanks to some innovative thinking and hard work, we have been able to navigate all the challenges easily."

Future plans

The team at IdeaForge says that the product has been commercialised and that they have already started selling Netra to various law enforcement and defence agencies. When asked about their plans to increase the features of the UAV, Vipul Joshi, COO, responded, "We keep improving the product at every opportunity. We have improved its flight time from 25 minutes plus to 30 minutes plus, its range from 1.5 km to 2.5 km, and so on. We are planning to enhance these basic parameters even further as well as offer a host of new image processing features in the next *avatars* of the UAV. We work very hard here to make sure this idea gets as awesome as we hope." Certainly, EFY readers would wish them the same. ●

The author is a tech correspondent at EFY Bengaluru

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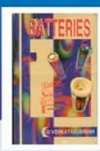
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